

Question 1

Reagent

Labelled	contents	hazard	Volume/cm ³
E1	Amylase solution	Irritant Harmful	5
E2	Amylase solution	Irritant Harmful	5
S	Starch solution	None	50
G1	0.4% reducing sugar solution	None	50
W	Distilled water	none	100
Benedict's solution	Benedict's solution	Irritant Harmful	100

- **E1** – at least 5 cm³ of 1% amylase solution in a container, labelled E1
E2 – at least 5 cm³ of 0.1% amylase solution in a container, labelled E2
Test the activity of the enzyme by putting drops of iodine solution onto a spotting tile and adding drops of E1 mixed with S at intervals. The colour should stay yellow-brown within 5 minutes.
- Starch solution, **S**, at least 50 cm³ of 1% starch solution in a beaker or container, labelled S. Prepared by putting 1 g of starch into 25 cm³ of warm distilled water in a beaker or container and mixing to a paste. Make up to 100 cm³ by adding boiling distilled water, mix well and then allow to cool.
- 0.4% glucose solution, **G**, at least 50 cm³ of 0.4% glucose solution in a beaker or container, labelled G. This is prepared by sprinkling 0.4 g of anhydrous glucose onto the surface of 80 cm³ of distilled water, stirring continuously as you sprinkle. Make up to 100 cm³ with distilled water.
- **W**, at least 100 cm³ of distilled water in a beaker or container, labelled **W**.
- Fresh Benedict's solution, at least 50 cm³ in a beaker or container, labelled Benedict's solution. This must be qualitative Benedict's solution as provided by your supplier or made up by the technician. Please note it is important to use hydrated copper sulfate crystals copper(II) sulfate(VI)-5-water when making up the Benedict's solution.

Question 1 (continued)

S/N	Apparatus for each candidate	Quantity
1	50 cm ³ beakers	6
2	500 cm ³ beakers	1
3	5 cm ³ syringe	2
4	15 cm ³ syringe	2
5	Test tubes	6
6	Test tube rack	1
7	Stop watch	1
8	Marker Pen	1
9	Safety goggles	1
10	Hot plate	1
11	Thermometer	1
12	Glove	1

Question 2

Reagent

Labelled	contents	hazard	Volume/cm ³
L1	Fresh leaf (spinach)	None	-
L2	Cold buffer solution	None	10
L3	Sand with spatula	None	20
L4	0.05% DCPIP	None	5

- Spinach leaf should be about 10cm x 4cm.
- Cold buffer solution, labelled L2 and supplied with a pasteur pipette. The solution should be presented to the candidate in a test tube stood in a beaker of ice cold water. This solution, L2, is made by making 6 g of sucrose and 0.01 g of KCl up to 100 cm³ with distilled water.
- 5cm³ of 0.05% DCPIP solution. The solution can be made by dissolving 0.5 g of DCPIP up to 1dm³ in distilled water labelled L4 and supplied with a teat pipette. It should also be presented to the candidate in a test tube stood in the same beaker of ice cold water as the buffer solution.

S/N	Apparatus for each candidate	Quantity
1	White tile	3
2	Scissor	1
3	Small plastic container	1
4	Pasteur pipette	1
5	Glass rod	1
6	Petri dish (without cover)	1
7	Aluminium foil (30cm x 30cm)	1
8	Capillary tubes	3
9	Bench lamp	1
10	Stop watch from question 1	1

Question 3

- Solanum T/S stem tuber to show starch grains labelled as **S1**
- Microscope
- Stage micrometer